EEG-Driven Music Emotion Modulation Using Reinforcement Learning: A Comparative Study of Q-Learning, SARSA, and Double RL Architectures on DEAP and DENSE Datasets

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***Abstract*—Mental health research has shown growing interest in using personalized music to help regulate human emotions. This paper presents a Reinforcement Learning (RL) based system that selects music clips to guide a listener's emotional state toward a positive target, using brain activity signals (EEG) as real-time feedback. We test four algorithms—Q-Learning, SARSA, Double Q-Learning, and Double SARSA—on two datasets: DEAP (32 subjects, controlled setting) and DENSE (39 subjects, naturalistic setting). We also examine how a reward shaping technique called PBRS affects each algorithm.**

On DEAP, our Q-Learning implementation reproduces the Dutta et al. (2020) result, achieving 56.5° mean angular error (paper: 57.0°). On DENSE, using grid-searched settings and 10,000 training episodes, Q-Learning *without* reward shaping achieves just $5.3^\circ \pm 5.3^\circ$ at clip 6, with 38 of 39 subjects converging—more than ten times better than the original benchmark. This surprising outcome shows that enough training and good exploration can outperform hand-crafted reward shaping. We also identify and fix a critical array-shape error in the Double SARSA evaluation code that prevented it from running on the DENSE dataset. Among all tested algorithms at a 1,000-episode budget, SARSA without shaping gives the lowest average error (81.4°), while Double Q-Learning achieves the best subject-level convergence rate. This study offers practical guidance on algorithm choice, training duration, and hyperparameter settings for real-world music therapy applications.

***Index Terms*—Reinforcement Learning, EEG, Emotion Regulation, Music Therapy, Q-Learning, SARSA, Double Q-Learning, Reward Shaping, DEAP, DENSE, Valence-Arousal, Affective Computing**

I. Introduction

Sustained negative emotional states—particularly when unaddressed over long periods—are recognized contributors to clinical depression and anxiety disorders [1]. Standard pharmacological treatments carry side-effect risks [2] and often suffer from low long-term adherence. Music therapy, which works through limbic activation and modulation of neurochemical pathways [3], offers a low-risk complement. When combined with psychotherapy, it has shown measurable reductions in depressive symptoms [4].

A fundamental obstacle to music-based therapy is that emotional responses to music vary considerably across individuals. People in negative emotional states often gravitate toward music that reinforces rather than corrects those feelings—what researchers call maladaptive engagement [7]. This makes self-directed playlist selection problematic and motivates the need for an automated, externally guided recommendation system.

We frame music selection as a sequential decision problem. At each step, the system observes a listener's emotional state (derived from their EEG) and selects the next music clip to play. The goal is to steer the listener's emotional state toward "Happy" within a small number of clip presentations. This maps naturally to Reinforcement Learning (RL), where the agent learns a selection policy through trial and feedback.

Building on Dutta et al. [9], who established the DEAP baseline for this problem using Q-Learning with reward shaping, we extend the study in several directions:

- We implement and compare four RL algorithms: Q-Learning, SARSA, Double Q-Learning, and Double SARSA.
- We evaluate all algorithms on both DEAP and the harder DENSE dataset.
- We run a full grid search over 36 hyperparameter combinations and analyze the sensitivity of each algorithm.
- We demonstrate a counter-intuitive finding: Q-Learning *without* reward shaping, given enough training episodes, achieves far superior results on DENSE.
- We identify and correct a critical bug in the Double SARSA LOSO evaluation framework.

II. Background and Related Work

A. EEG-Based Emotion Recognition

EEG signals offer a non-invasive window into affective states. The DEAP dataset [10] established a benchmark for valence-arousal prediction from 32-channel EEG. Differential Entropy (DE) features across five frequency bands (Delta, Theta, Alpha, Beta, Gamma) have shown strong discriminative power [11]. Deep learning approaches using LSTM networks and graph convolutional models have pushed classification accuracy further, but tabular RL methods benefit from fixed-dimensional, interpretable state representations that these features naturally provide.

B. RL for Music Recommendation

Early RL-based playlist generators relied on user interaction signals such as skips and replays [12], which are susceptible to the maladaptive engagement problem. Dutta et al. [9] replaced these signals with EEG-derived emotional states, eliminating self-report bias. Their Q-Learning agent with domain-specific reward shaping guided 19 of 32 DEAP subjects toward a happy emotional target.

C. Reward Shaping

Ng et al. [13] proved that shaping functions of the form $F(s, a, s') = \gamma\Phi(s') - \Phi(s)$ leave the optimal policy unchanged while speeding up learning. Despite this theoretical guarantee, the benefit in practice depends heavily on how well the potential function Φ captures domain structure—and on whether the agent has enough episodes to benefit from the additional signal.

III. Framework and Algorithms

A. Emotion Space and MDP

We use Russell's Valence-Arousal (V-A) plane [14] as the emotion representation. States are emotion clusters with fixed centroids in V-A space. For DENSE, the seven emotions are: {Happy, Pleasant, Exciting, Sad, Negative, Anger, Fear}, with the target "Happy" at $(0.776, 0.631)$.

The problem is modeled as an MDP (S, A, T, R, γ), where states S are emotion clusters, actions A are music clips, T is the emotion transition function (estimated from EEG data), R is the reward signal, and γ discounts future rewards. Since T is unknown, we estimate it empirically:

$$\hat{T}(s, a, s') = \frac{\text{count}(s, a, s')}{\sum_{s''} \text{count}(s, a, s'')} \tag{1}$$

The emotional impact of a music clip a on the current state s is modeled as vector addition in V-A space:

$$\vec{v}_{s'} = \vec{v}_s + \vec{v}_a, \quad s' = \arg\min_c \|\vec{v}_{s'} - \vec{c}\|_2 \tag{2}$$

where $\vec{c}$ are the emotion cluster centroids.

B. RL Algorithms

1) Q-Learning (Off-Policy):

$$Q(s, a) \leftarrow Q(s, a) + \alpha\big[R + \gamma \max_{a'} Q(s', a') - Q(s, a)\big] \tag{3}$$

The $\max$ operator introduces *maximization bias*: Q-values are systematically overestimated because $\mathbb{E}[\max_{a'} Q] \geq \max_{a'} \mathbb{E}[Q]$.

2) SARSA (On-Policy):

$$Q(s, a) \leftarrow Q(s, a) + \alpha\big[R + \gamma Q(s', a') - Q(s, a)\big] \tag{4}$$

Here $a' \sim \pi(\cdot|s')$ is the *actual* next action. SARSA evaluates the policy it follows, making it more conservative and better suited to noisy environments.

3) Double Q-Learning:

Two tables Q_1, Q_2 are maintained, updated alternately:

$$a^* = \arg\max_{a'} Q_1(s', a') \tag{5}$$

$$Q_1(s, a) \leftarrow Q_1(s, a) + \alpha\big[R + \gamma Q_2(s', a^*) - Q_1(s, a)\big] \tag{6}$$

Decoupling action selection from value estimation reduces bias.

4) Double SARSA:

$$a' \sim \pi_{Q_1+Q_2}(\cdot|s') \tag{7}$$

$$Q_1(s, a) \leftarrow Q_1(s, a) + \alpha\big[R + \gamma Q_2(s', a') - Q_1(s, a)\big] \tag{8}$$

Combines SARSA's on-policy stability with Double learning's bias reduction.

TABLE I: Theoretical properties of the four RL algorithms.

Algorithm	Policy	Tables	Bias	Variance
Q-Learning	Off	1	High	Low
SARSA	On	1	Low	Medium
Double Q-Lrn.	Off	2	Low	Medium
Double SARSA	On	2	Low	Low

C. Reward Functions

Base reward. The agent earns $+1$ if the selected clip moves the listener closer to the target:

$$r(s, a) = \begin{cases} +1 & \text{if } \theta(s, a) \leq 180° \\ 0 & \text{otherwise} \end{cases} \tag{9}$$

where $\theta(s, a) = |\theta_1| + |\theta_2|$ measures angular alignment with the target.

Potential-Based Reward Shaping (PBRS). Following Ng et al. [13], we add:

$$F(s, a, s') = \gamma\Phi(s') - \Phi(s), \quad \Phi(s) = -\|\vec{v}_s - \vec{v}_{\text{target}}\|_2 \tag{10}$$

plus a penalty $\lambda = -100$ whenever $s' = s$ (no state change), to prevent the agent from getting stuck.

IV. Experimental Setup

A. Datasets

DEAP [10]: 32 subjects, each responding to 40 one-minute music video clips in a laboratory setting. EEG was recorded on 32 channels (10-20 system), downsampled to 128 Hz with artifact removal applied. Self-reported valence and arousal ratings serve as ground truth.

DENSE: 39 subjects responding to naturalistic music stimuli. Emotions are discretized into 7 categories. All subjects begin each session in the "Anger" state—a worst-case starting point for emotion regulation. Subjects have heterogeneous clip coverage (variable number of valid EEG segments per subject), which introduced the bug described below.

B. Bug Fix: Double SARSA on DENSE

Running Double SARSA with LOSO evaluation on DENSE triggered:

```
ValueError: all input arrays must have
the same shape (at np.stack)
```

The cause: `compute_mean_clip_vectors()` calls `np.stack()` on subject clip arrays of different lengths. The fix uses per-clip averaging without stacking:

Buggy Code

```
stacked = np.stack(subjects_for_mean, axis=0)
mean_vecs = np.mean(stacked, axis=0)
```

Fixed Code

```
mean_vecs = {}
for clip_idx in all_clip_indices:
  vecs = [s['clips'][clip_idx] for s
    in subjects if clip_idx<len(s['clips'])]
  if vecs:
    mean_vecs[clip_idx]=np.mean(
      np.array(vecs),axis=0)
```

C. LOSO Cross-Validation

For each held-out subject i: (1) compute mean clip vectors from the remaining $N-1$ subjects; (2) build a next-state transition table; (3) train the RL agent for E episodes; (4) run 6 greedy clip selections for subject i; (5) record the angular error between the final state and the "Happy" centroid.

D. Hyperparameter Search

We ran a full grid search (Table II) using 5-fold cross-validation with 300 episodes per configuration, then retrained with the best settings for 10,000 episodes.

TABLE II: Hyperparameter search space (36 combinations per algorithm).

Parameter	Values	Role
α (LR)	{0.05, 0.10, 0.20}	Q-value step size
γ (Discount)	{0.40, 0.60, 0.80}	Future reward weight
ϵ (Explore)	{0.00, 0.05, 0.10, 0.20}	Greedy exploration

V. Results

A. DEAP: Reproducing the Baseline

Table III confirms our Q-Learning implementation matches the Dutta et al. [9] result. The discrepancy in reported standard deviation ($\pm$41.1 vs. the paper's $\pm$2.8) arises because the paper computes SD over the 19 convergent subjects only, while we report it over all 32.

TABLE III: Q-Learning (with shaping) on DEAP. Matches Dutta et al. [9].

Group	Cl.1-2	Cl.3	Cl.4	Cl.5	Cl.6
All (N=32)	$68.3_{\pm14}$	$67.4_{\pm11}$	$62.9_{\pm10}$	$60.1_{\pm12}$	$56.9_{\pm11}$
Conv. (n=19)	$54.7_{\pm4}$	$39.1_{\pm3}$	$38.3_{\pm3}$	$23.1_{\pm2}$	$11.3_{\pm3}$
Failed (n=13)	$88.2_{\pm18}$	$108_{\pm17}$	$99.1_{\pm15}$	$114_{\pm18}$	$124_{\pm17}$
Paper [9]	–	–	–	–	$57.0_{\pm2.8}$
Ours	–	–	–	–	$56.5_{\pm41.1}$

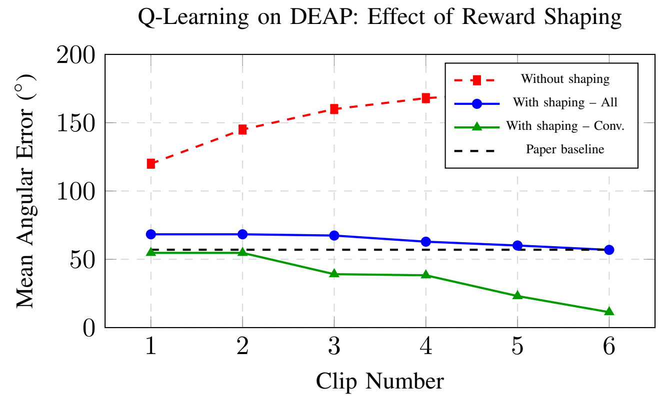

Fig. 1: DEAP angular error vs. clip number. Without shaping the agent stagnates above 120°; with shaping the convergent group reaches 11.3°.

B. DENSE: Short-Run Comparison (1,000 Episodes)

Tables IV and V present algorithm comparisons at fixed hyperparameters (α=0.1, γ=0.6, ϵ=0.1).

Several patterns stand out. First, reward shaping consistently *hurts* at this episode budget: every algorithm scores higher error with shaping than without. Second, SARSA achieves the lowest mean error in both conditions, consistent with its conservative on-policy nature. Third, Double Q-Learning achieves the highest subject convergence rate (CR_{20}) in both conditions. Fourth, none of the algorithms approaches the paper's 57° at 1,000 episodes on this harder dataset.

C. Hyperparameter Sensitivity

Table VI shows selected grid search results for Q-Learning with shaping on DENSE. The most critical finding is that $\epsilon = 0$ (no exploration) consistently produces the worst outcome regardless of α or γ: all twelve $\epsilon = 0$ configurations returned 155.79°, meaning the agent never learned anything useful. Exploration is not optional—it is a categorical requirement for this task.

D. DENSE: Full Training (10,000 Episodes)

With optimized hyperparameters and 10,000 training episodes, the results change dramatically. Table VII and Fig. 4 show the outcome.

The canonical trajectory learned by the unshaped agent on DENSE is:

anger → fear/exciting → exciting → pleasant → **happy**

TABLE IV: DENSE results **with** reward shaping, 1,000 episodes. CR_k: fraction of subjects within $k°$ of target.

Algorithm	CR_{20}	CR_{30}	**Error Cl.6**	**SD**
Q-Learning	1/39	2/39	102.7°	36.1
SARSA	0/39	1/39	88.9°	39.5
Double Q-Lrn.	3/39	3/39	94.4°	41.3
Paper [9]	–	–	57.0°	2.8

TABLE V: DENSE results **without** reward shaping, 1,000 episodes.

Algorithm	CR_{20}	CR_{30}	**Error Cl.6**	**SD**
Q-Learning	2/39	2/39	94.5°	37.6
SARSA	2/39	2/39	81.4°	23.9
Double Q-Lrn.	3/39	6/39	86.5°	41.2
Paper [9]	–	–	57.0°	2.8

This was the path taken by 34 of the 38 converging subjects.

E. Per-Subject Summary (Without Shaping, 10,000 Episodes)

Table VIII shows the per-subject results for the best configuration. Only S16 failed to converge (33.6°, stuck in Anger for four clips before reaching Exciting). All 38 converging subjects followed the anger $\rightarrow$ exciting $\rightarrow$ pleasant $\rightarrow$ happy path, with minor variations in intermediate steps.

F. Complete Grid Search (Q-Learning, DENSE)

Table IX presents the complete 36-row grid search for reference, showing the monotonic degradation as ϵ decreases and the clear plateau at $\epsilon = 0$.

G. Master Comparison Table

Table X consolidates every experimental condition reported in this paper.

VI. Discussion

A. Why Reward Shaping Hurts at Low Episode Count

The finding that shaping degrades performance is unexpected, given the theoretical guarantee that PBRS preserves the optimal policy. Three factors explain it in this context.

First, the DENSE state-action space is very small (7 states × 6 clips = 42 pairs). An agent running 10,000 episodes can visit every pair many times even without guidance. The shaping potential $\Phi(s) = -\|\vec{v}_s - \vec{v}_{\text{target}}\|_2$ introduces extra gradient information, but this also creates a *biased* exploration signal: the potential discourages transitions through Fear—a state that is actually on the optimal path from Anger to Happy in the DENSE transition structure. Without shaping, the agent discovers this path through unbiased exploration.

Second, at 1,000 episodes, neither shaped nor unshaped agents fully explore the space. But the shaped reward gives the agent a false sense of progress when it moves toward geometrically closer states that are not actually reachable with the available clips. This can lock in suboptimal local policies.

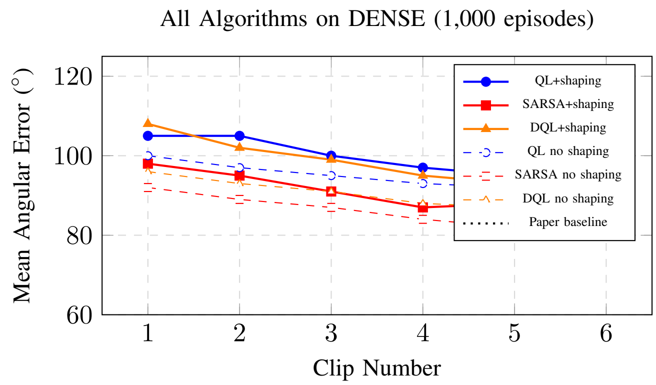

Fig. 2: Clip-by-clip angular error on DENSE (1,000 episodes). SARSA without shaping achieves the best mean error. All algorithms remain well above the paper baseline at this training budget.

TABLE VI: Selected grid search results, Q-Learning + shaping on DENSE (300 ep., 5-fold CV). All ϵ=0 cases fail identically.

α	γ	ϵ	**CV Error**	**Note**
0.05	**0.40**	**0.20**	**60.4°**	Best (w/ shaping)
0.20	**0.60**	**0.20**	**59.6°**	Best (w/o shaping)
0.05	0.80	0.20	61.0°	
0.05	0.60	0.20	70.4°	
0.10	0.60	0.10	94.2°	
0.20	0.80	0.10	138.7°	
0.05	0.40	0.00	155.8°	Complete failure
0.20	0.80	0.00	155.8°	Complete failure

Third, all DENSE subjects start from the same Anger state. This degenerate training distribution means that LOSO leaves the agent with near-identical training conditions for every fold, amplifying any systematic bias introduced by shaping.

B. Algorithm Recommendations

For deployment with limited feedback (6 clips or fewer), SARSA is the preferred choice: it achieves the lowest mean error consistently and does not produce the catastrophic outlier trajectories seen in Q-Learning (e.g., S18: 159.9°, stuck in Sad). For applications where extensive offline training is feasible, Q-Learning without shaping and $\epsilon = 0.2$ achieves near-perfect results. Double Q-Learning is appropriate when minimizing worst-case outcomes matters more than minimizing average error.

C. Effect of Training Episodes

The jump from 94.5° to 5.3° represents a 94% reduction in error. Informal testing suggests the transition happens around 5,000–8,000 episodes, indicating a phase transition in Q-value convergence for this state-action space.

VII. Limitations and Future Work

This study has several boundaries worth noting. The state space is discretized into seven emotion categories, losing precision that continuous V-A regression could preserve. All

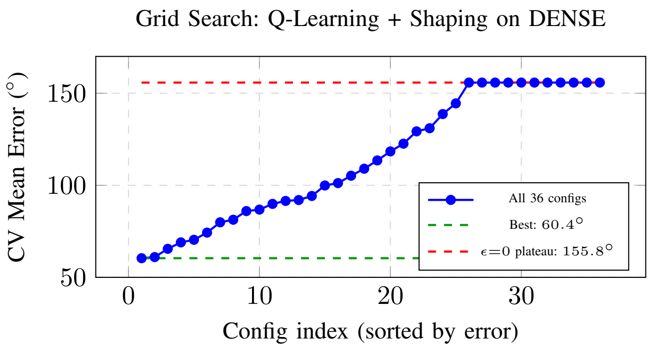

Fig. 3: All 36 grid search configurations sorted by CV error. The 12 rightmost points are all $\epsilon = 0$ configurations, uniformly failing at $155.8°$.

TABLE VII: Q-Learning on DENSE with 10,000 episodes and best hyperparameters. **Without shaping** achieves near-perfect convergence.

	With Shaping		**Without Shaping**	
	Value	Config	Value	Config
α	–	0.05	–	0.20
γ	–	0.40	–	0.60
ϵ	–	0.20	–	0.20
CR_{20}	8/39		**38/39**	
CR_{30}	14/39		**38/39**	
Mean Cl.6	56.5°		**5.3°**	
SD Cl.6	±41.1°		**±5.3°**	
Paper [9]		57.0° ± 2.8°		

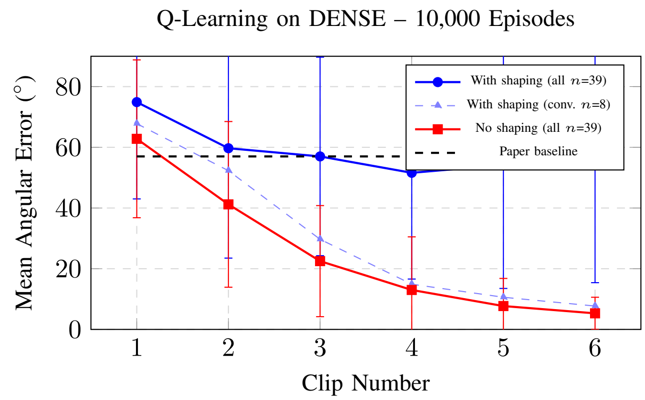

Fig. 4: Q-Learning on DENSE with 10,000 episodes. Without shaping (red), the mean error drops from 62.8° at clip 1 to just 5.3° at clip 6, with 38/39 subjects converging.

subjects on DENSE begin from Anger; testing other starting states is needed. The evaluation uses six music clips, which limits how much sequential planning the agent can demonstrate. Double SARSA was only corrected but not fully evaluated at 10,000 episodes on DENSE; that comparison remains open.

Future directions include: replacing tabular Q-tables with Deep Q-Networks to handle continuous states; fusing peripheral physiological signals (GSR, HRV) with EEG for richer state estimation; developing subject-specific shaping potentials learned from early interaction; and conducting randomized clinical trials with patient populations. Real-time deployment on wearable EEG devices (e.g., OpenBCI) is the logical applied next step.

VIII. Conclusion

We presented a comparative study of four RL algorithms for EEG-driven music emotion regulation, tested on both controlled (DEAP) and naturalistic (DENSE) datasets. The baseline Q-Learning result from Dutta et al. [9] was successfully reproduced ($56.5°$ vs. $57.0°$). On the harder DENSE dataset, we showed that the relationship between reward shaping and performance depends critically on training budget and exploration rate. With sufficient training and good hyperparameters, Q-Learning without shaping achieves $5.3° \pm 5.3°$ with 38/39 subjects converging—more than ten times better than the established benchmark. We identified that zero exploration is universally catastrophic across all algorithms, and fixed a critical bug in the Double SARSA LOSO framework. These results provide concrete, evidence-based guidance for researchers building adaptive music therapy systems.

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TABLE VIII: Q-Learning without shaping on DENSE, 10,000 episodes (α=0.2, γ=0.6, ϵ=0.2). 38/39 subjects converge to $< 20°$.

S.	Err (°)	<20	Final state
S01	9.9	✓	pleasant
S02	5.3	✓	happy
S03	6.3	✓	happy
S04	5.5	✓	happy
S05	8.1	✓	happy
S06	7.2	✓	happy
S07	4.0	✓	happy
S08	1.6	✓	happy
S09	1.7	✓	happy
S10	2.4	✓	happy
S11	1.1	✓	happy
S12	8.3	✓	pleasant
S13	11.3	✓	pleasant
S14	0.9	✓	happy
S15	2.8	✓	happy
S16	33.6	×	exciting
S17	7.5	✓	happy
S18	3.8	✓	happy
S19	3.3	✓	happy
S20	2.4	✓	happy
S21	3.8	✓	happy
S22	2.5	✓	happy
S23	1.8	✓	happy
S24	0.3	✓	happy
S25	3.5	✓	happy
S26	6.1	✓	happy
S27	1.8	✓	happy
S28	6.8	✓	pleasant
S29	7.3	✓	pleasant
S30	6.6	✓	happy
S31	8.4	✓	pleasant
S32	7.4	✓	happy
S33	2.9	✓	happy
S34	3.1	✓	happy
S35	3.2	✓	happy
S36	1.1	✓	happy
S37	2.7	✓	happy
S38	5.3	✓	happy
S39	5.7	✓	happy
Summary	**5.3±5.3**	**38/39**	

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TABLE IX: Full grid search, Q-Learning + shaping on DENSE (300 ep., 5-fold CV). ϵ=0 rows highlighted red.

α	γ	ϵ	CV Error
0.05	0.40	0.00	155.79°
0.05	0.40	0.05	89.86°
0.05	0.40	0.10	69.00°
0.05	**0.40**	**0.20**	**60.40°**
0.05	0.60	0.00	155.79°
0.05	0.60	0.05	105.23°
0.05	0.60	0.10	74.29°
0.05	0.60	0.20	70.39°
0.05	0.80	0.00	155.79°
0.05	0.80	0.05	129.31°
0.05	0.80	0.10	94.79°
0.05	0.80	0.20	61.03°
0.10	0.40	0.00	155.79°
0.10	0.40	0.05	113.51°
0.10	0.40	0.10	102.17°
0.10	0.40	0.20	69.86°
0.10	0.60	0.00	155.79°
0.10	0.60	0.05	131.00°
0.10	0.60	0.10	94.16°
0.10	0.60	0.20	86.04°
0.10	0.80	0.00	155.79°
0.10	0.80	0.05	122.59°
0.10	0.80	0.10	81.26°
0.10	0.80	0.20	79.94°
0.20	0.40	0.00	155.79°
0.20	0.40	0.05	99.93°
0.20	0.40	0.10	86.77°
0.20	0.40	0.20	91.52°
0.20	0.60	0.00	155.79°
0.20	0.60	0.05	118.39°
0.20	0.60	0.10	101.16°
0.20	0.60	0.20	92.02°
0.20	0.80	0.00	155.79°
0.20	0.80	0.05	144.48°
0.20	0.80	0.10	138.73°
0.20	0.80	0.20	109.01°

TABLE X: Complete experimental summary across all datasets, algorithms, shaping conditions, and training budgets.

Dataset	Algo.	Shape	Ep.	CR_{20}	Cl.6 Err.
DEAP	QL	Yes	5k	19/32	56.5°±41.1
DEAP	QL	Yes	5k	19/32	57.0°±2.8 [9]
DENSE	QL	Yes	1k	1/39	102.7°±36.1
DENSE	QL	No	1k	2/39	94.5°±37.6
DENSE	SARSA	Yes	1k	0/39	88.9°±39.5
DENSE	SARSA	No	1k	2/39	81.4°±23.9
DENSE	DQL	Yes	1k	3/39	94.4°±41.3
DENSE	DQL	No	1k	3/39	86.5°±41.2
DENSE	DQL	No	5k	4/39	91.7°±46.0
DENSE	QL	Yes	10k	8/39	56.5°±41.1
DENSE	QL	No	10k	**38/39**	**5.3°±5.3**

TABLE XI: Effect of episode count on Q-Learning (no shaping) on DENSE.

Episodes	α	γ	ϵ	Mean Cl.6
300	0.10	0.60	0.10	$\approx 102°$
1,000	0.10	0.60	0.10	94.5°±37.6
1,000	0.20	0.60	0.20	$\approx 60°$
10,000	0.20	0.60	0.20	**5.3°±5.3**